

Hidden Effects of Feature Scaling Choices on Logistic Regression Models

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Abstract

Logistic regression models are great supervised learning tools to classify data across multiple domains such as finance and healthcare. Other researchers have studied the effectiveness of scaling on these models but have concluded that it is widely dataset dependent. This study explores the underlying effects of standardization min max scaling and what attributes contribute to these differences. To do this, we found three datasets from different domains (Heart Disease, raisins, loans) and ran 3 trials of logistic regression models on each scaling type. We compared the results of each model noting accuracy, precision, recall, and F1-score. What we found was that in datasets with high magnitude difference between features, there was an increase in all metrics while the difference was not significant in other datasets, with some metrics even decreasing. We conclude that scaling is important for less uniform data and provides marginal benefit for uniform data. For future work, we suggest exploring the quantitative relationship between feature distribution and the effectiveness of scaling techniques.

1 Introduction

Within the field of Artificial Intelligence (AI), there is a subfield known as Machine Learning (ML) which is concerned with predicting future outcomes based on data from the past. There are many different machine learning techniques used for specific tasks, these are known as models. Among these, logistic regression is a well known and powerful machine learning model for supervised learning, specifically for classification problems. Its goal is to predict the probability of a categorical value. Due to its simplicity and efficiency, it is a strong candidate for fields such as healthcare and finances.

Logistic regression models are commonly evaluated on testing metrics such as accuracy, precision, recall and F1-score. The accuracy of a model refers to the proportion of correctly classified instances relative by the total number of instances in the dataset. Precision measures the proportion of instances predicted as positive that are truly positive.

Recall calculates the proportion of actual positive instances that were correctly identified by the model. Finally, the F1-Score looks to combine recall and precision measures (Lasisi and Lasisi 2025).

Machine Learning models can be influenced by how data is preprocessed prior to training. For logistic regression models, an important aspect of data preprocessing is feature scaling, which consists in transforming variables so they can share similar ranges and distribution. Common feature scaling techniques include standardization and normalization. The extend to which logistic regression models benefit from feature scaling remains an open question, some studies claim significant improvement on model performance, while some claim that it dataset dependent.

This paper aims to investigate the impact of feature scaling techniques on logistic regression models. By combining a systematic literature review with experimentation across multiple datasets, we intend to learn about the hidden effects of feature scaling choice, specifically unscaled data, standardization, and Min-Max scaling.

2 Literature Review

Extensive research has been conducted to determine the effects of feature scaling on logistic regression models.

In (Herrera et al. 2025) the authors evaluated how different regression and classification models react to diverse feature scaling techniques across different datasets. For our particular study we focused on the results for logistic regression models when faced against different scaling techniques. The authors demonstrate that logistic regression models are highly sensitive to feature scaling, with the choice of data transformation drastically influencing accuracy and training resources. In this study, it was not possible to pinpoint a single scaling technique that would always positively influence the logistic regression model. The authors concluded that feature scaling techniques must be carefully selected based on the characteristics of the dataset.

In (Ozsahin et al. 2022) the authors evaluated how

normalization and standardization affected logistic regression models when it came to diagnosing diabetes. The model reached its peak accuracy of 75 percent when standardization and no feature scaling was applied. When normalization was applied, the model performance actually decreased to 74 percent. These results led the researchers to conclude that feature scaling has no significant impact on the logistic regression model for this specific dataset.

In (Mathivanan et al. 2025) the authors explore how feature standardization can have an effect on the heart disease prediction capabilities of a logistic regression model. The dataset utilized in this study is composed of demographic and clinical attributes of patients. F1-scores for logistic regression models increased from 0.82 to 0.87 when standardization was applied. Similarly, hyperparameter values increased from 0.001 to 0.01 when standardization was applied. Based on these performance metrics, it is clear that logistic regression models are impacted by standardization for this particular dataset. These findings highlight the importance of data pre-processing.

In (Ashan et al. 2021) a University of California heart disease data set was used to measure the effects of different scaling methods on 11 different machine learning algorithms, including a logistic regression model. The scaling methods used were Normalization, StandardScaler, Min-Max, MaxAbs, RobustScaler, and Quantile Transformer. The results showed consistent levels of accuracy across the scaling methods with the exception of min max dropping by 0.2, and precision within 0.01 for most methods with the exception of Normalization being significantly worse. In general, the accuracy of these scaling methods had very little positive impact on the logistic regression model. What this paper fails to explore are multiple datasets with different domains. This is something we will try in our paper.

In (Oka 2021) a study was conducted that uses a dataset on public health and socioeconomic status and multilevel logistic regression to show how the same data can produce different interpretations of coefficients depending on whether standardization or normalization is applied. The results of this study show that the resulting Odds Ratios in these models are derived directly from the magnitudes of scaling. This article is the foundation for our assumption going into our study that standardization is required for complete transparency, because without it the odds ratio is an arbitrary number that makes it impossible to come to definitive conclusions.

In (Sujan et al. 2022) multiple datasets from different domains were used to test the effects of Standardization, Normalization, and Unscaled Data across SVM, Logistic Regression, Random Forest, XGBoost, and AdaBoost. It is important to note that these datasets ranged in size, making for a well rounded study. They found that Logistic regression showed improvements on accuracy with standardization in medium to large datasets, while small

datasets were improved with normalization. Our study will directly seek to verify these findings and improve on their procedure to focus solely on the logistic regression model in depth.

3 Research Methodology

A typical preprocessing of a dataset consists in cleaning it from any noise introduced at the time of collection. This noise can be in the form of duplicate data, inconsistent values, missing values, or encoding of categorical data. However, depending on the magnitudes between features, this could raise a question if the magnitudes of the features need to be leveled using techniques such as standardization or MinMax scaling, or just continue with the model training on the raw data. This study aims to determine the hidden effects of scaling features in the training of logistic regression models and see whether it benefits the model's performance or not.

This study consists of two components: A systematic review of peer reviewed articles related to the effects of feature scaling in logistic regression models, and a model training procedure utilizing Google Colab environment to test the performance on our models across different feature scaling techniques to determine which are more beneficial during the data preprocessing.

Original datasets used to train our models can be publicly found on Kaggle following these links:

Link to Dataset 1 - Bank Loan Approval.

Link to Dataset 2 - Raisin Type Classification.

Link to Dataset 3 - Heart Disease Cleveland.

Our first dataset consisted of a bank's customer loan requests. The goal was to predict if the loan was approved based on features such as age, annual income, credit score, loan amount, and savings assets. This dataset is interesting because values for different features vary widely, ranging from 0 up to 300000. To address this variation, standardization and Min-Max feature scaling were applied to columns with values in the hundreds or higher, while columns such as loan to income ratio were left unchanged since their values ranged only from 0 to 2. This preprocessing step helped to ensure that large scaled features did not dominate small scaled features.

Our second dataset consisted of classifying raisins into either Kecimen or Besni based on features including Area, Major Axis Length, Minor Axis Length, Eccentricity, Convex Area, Extent, and Perimeter. The values for this dataset ranged from 0 to 278217, although the majority of the values observed were under 1000. Similarly to the first dataset, standardization and Min-Max scaling were applied to columns with larger numerical ranges, while columns with maximum values of 1 were left as is.

Our third and last dataset consisted of diagnosing patients with heart disease or not based on factors such as age,

sex, gender, chest pain type, blood pressure, cholesterol, blood sugar levels, ecg results, maximum heart rate, angina (boolean), number of major vessels. Similarly to the first and second datasets, standardization and Min-Max scaling were applied to columns with larger numerical ranges, while columns with maximum values of 1 were left as is. This approach maintained consistency in preprocessing across datasets.

After loading and cleaning our initial dataset in the Google Colab environment, we trained and tested our model on the unscaled data. This allowed us to establish a baseline performance on raw data to better understand the impact of feature scaling. The first feature scaling technique applied was standardization, which transforms the variables to have a mean of 0 and a standard deviation of 1. This is achieved by subtracting the mean of the variable from each data point and then dividing it by the standard deviation of the variable (Lasisi and Lasisi 2025). We then repeated the process using Min-Max scaling which transforms the data points into a specified interval of range 0-1. This is achieved by subtracting the minimum value from each data point and then dividing it by the variable range which is defined as the difference between the maximum and minimum values (Lasisi and Lasisi 2025). Finally we compared the performance of all three models across metrics such as accuracy, precision, recall, and F1 score to determine the impact of feature scaling to the dataset and see if of one technique proved to be superior to the other or if they both performed the same.

4 Results and Discussions

The performance of the logistic regression model was evaluated across three distinct preprocessing settings: unscaled (raw) data, standardization, and Min-Max scaling. The results for the first dataset are summarized in Table 1:

Metric	No Scaling	Standardization	Min-Max
Accuracy	0.74	0.87	0.87
Precision	0.75	0.87	0.87
Recall	0.78	0.89	0.89
F1 Score	0.77	0.88	0.88

Table 1: Performances of models for the loan approval dataset with and without feature scaling

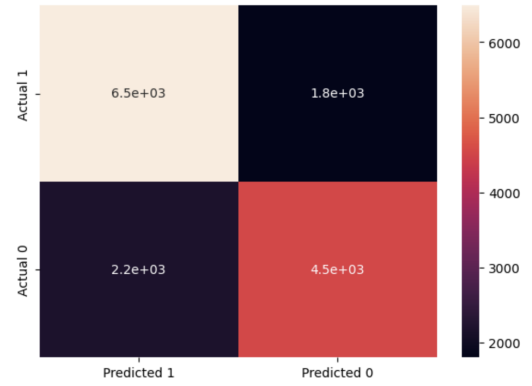


Figure 1: Bank loan approval confusion matrix using the raw model.

As shown in Table 1, the model presented significant performance increase when feature scaling was applied. Accuracy increased from 0.74 with no feature scaling to 0.87 with both standardization and Min-Max scaling. Similarly, precision increased from 0.75 to 0.87, recall increased from 0.78 to 0.89, and F1-score increased from 0.77 to 0.88. Both feature scaling techniques had identical results across all metrics, suggesting that benefits come from balancing the values in the dataset instead of the specific scaling method used.

The results for the second dataset are shown in Table 2. In this case the model performed well without scaling. After applying standardization and Min-Max scaling, accuracy slightly increased from 0.89 to 0.90, precision increased from 0.87 to 0.88 with standardization, but decreased to 0.86 with Min-Max, recall increased from 0.91 to 0.92 and 0.93 respectively, and F1-score increased only with Min-Max scaling from 0.89 to 0.90.

Metric	No Scaling	Standardization	Min-Max
Accuracy	0.89	0.90	0.90
Precision	0.87	0.88	0.86
Recall	0.91	0.92	0.93
F1 Score	0.89	0.89	0.90

Table 2: Performances of models for the Raisin Classification dataset with and without feature scaling

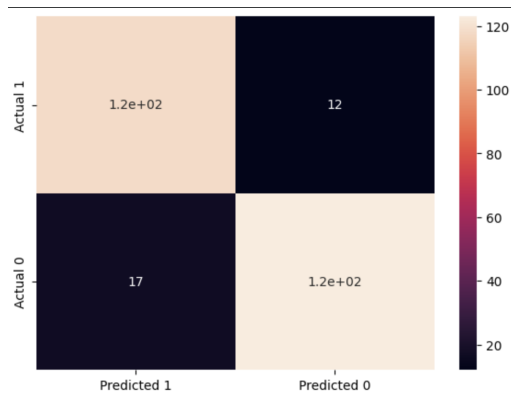


Figure 2: Raisin type confusion matrix using the raw model.

The results for the third dataset are shown in Table 3. In this case, the model performed well without scaling. After applying standardization, all our testing metrics increased by 0.01. However, after applying Min-Max scaling, only accuracy and precision increased by 0.1 and 0.3 respectively. F1-score remained the same, while the recall actually dropped from 0.85 to 0.83

Metric	No Scaling	Standardization	Min-Max
Accuracy	0.86	0.87	0.87
Precision	0.85	0.86	0.88
Recall	0.85	0.86	0.83
F1 Score	0.85	0.86	0.85

Table 3: Performances of models for the Heart disease Classification dataset with and without feature scaling

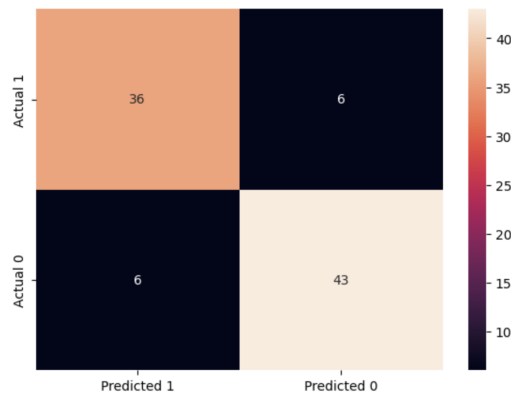


Figure 3: Heart disease Cleveland confusion matrix using the raw model.

Overall, as shown in the tables above, the models demonstrated consistent performance increase when feature scaling was applied, although the range of improvement was different in each dataset. For the loan approval dataset, feature scaling led to a significant improvement across all

testing metrics. In contrast, the raisin classification dataset barely showed any signs of improvement. This is likely because the majority of values were already in a similar range. The third dataset, Cleveland Heart disease, also saw a slight improvement from the scaling techniques used. This was likely because similar to the second dataset, the values were already in similar ranges.

Both standardization and Min-Max scaling were effective in this study, neither techniques showed clear advantages over the other, indication that both are suitable for improving logistic regression models in these contexts. It is important to note that only the Min-Max scaling technique had a few instances where it decreased the model's performance. This was observed in the raisin dataset's precision and in the Cleveland dataset's recall. However the decrease was only 0.1 and 0.2 respectively, which falls within the expected variability.

5 Conclusions and Future Work

This research investigated how feature scaling affects the accuracy, recall, precision, and F1-score of a logistic regression model. This was accomplished by first training a logistic regression model on raw data and then retraining it using both standardization and Min-Max scaling. This procedure was repeated across three datasets which included a bank loan approval, raisin classification, and a Cleveland heart disease dataset .

The results indicate that when applied correctly, both standardization and Min-Max scaling can improve model performance. The most significant improvement was observed in the bank loan approval dataset. This was due to the large difference in magnitude of the columns. For example, savings assets column contained values of hundreds of thousands while debt to income ratio did not surpass 0.800.

In contrast, the raisin classification and Cleveland heart disease datasets, only saw performance increases between 0.1 and 0.3. This was due to the reduced difference in magnitude for the columns. Contrary to the bank loan dataset, these contained the majority of values in the hundreds instead of hundreds of thousands.

For future work we suggest exploring the quantitative relationship between feature distribution and the effectiveness of scaling techniques. In our study, performance improvements were explained qualitatively (high vs low). However, a more rigorous approach can involve computing statistical measures for each feature in a dataset that would then correlate to a model's performance after applying scaling methods. This approach can allow researchers to better anticipate when feature scaling would provide significant improvements in model performance.

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